
Supplement: When Sharing Fails Under Massart Noise

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A External-Atom Moment Pairs

We give complete details for the moment construction. Let

$$d\sigma(u) = \frac{du}{\pi\sqrt{1-u^2}}, \quad u \in (-1, 1),$$

and let T_j be the degree- j Chebyshev polynomial of the first kind. The orthonormal polynomials in $L_2(\sigma)$ are

$$\phi_0(u) = 1, \quad \phi_j(u) = \sqrt{2}T_j(u), \quad j \geq 1.$$

For $m \geq 1$ and $z > 1$, define

$$K_m(z) = \sum_{j=0}^m \phi_j(z)^2 = 1 + 2 \sum_{j=1}^m T_j(z)^2.$$

Lemma 1 (External-atom representation). *There is a probability measure ν_m supported on $[-1, 1]$ such that*

$$\sigma_1 = K_m(z)^{-1} \delta_z + (1 - K_m(z)^{-1}) \nu_m$$

has the same moments as σ through order $2m$. The atom mass $K_m(z)^{-1}$ is maximal.

Proof. Let $w = K_m(z)^{-1}$. For a polynomial P of degree at most $2m$, define

$$L(P) = \int_{-1}^1 P(u) d\sigma(u) - wP(z).$$

We first show that $L(P) \geq 0$ whenever P is nonnegative on $[-1, 1]$. The univariate Markov–Lukacs representation gives

$$P(u) = s(u)^2 + (1 - u^2)t(u)^2$$

for polynomials satisfying $\deg(s) \leq m$ and $\deg(t) \leq m - 1$. This representation covers degree below $2m$ by allowing zero leading coefficients. See Wu and Yang (2020, Proposition 2.5.1).

Expand $s = \sum_{j=0}^m c_j \phi_j$. Cauchy–Schwarz gives

$$\begin{aligned} s(z)^2 &= \left(\sum_{j=0}^m c_j \phi_j(z) \right)^2 \\ &\leq \left(\sum_{j=0}^m c_j^2 \right) \left(\sum_{j=0}^m \phi_j(z)^2 \right) = K_m(z) \int s^2 d\sigma. \end{aligned} \tag{1}$$

Since $wK_m(z) = 1$ and $z > 1$,

$$\begin{aligned} L(P) &= \int s^2 d\sigma - ws(z)^2 + \int (1 - u^2)t(u)^2 d\sigma \\ &\quad + w(z^2 - 1)t(z)^2 \geq 0. \end{aligned}$$

The truncated Riesz–Haviland theorem (Wu and Yang, 2020, Theorem 2.5.2) now gives a nonnegative measure τ supported on $[-1, 1]$ such that

$$L(P) = \int P d\tau$$

for every polynomial of degree at most $2m$. Taking $P = 1$ shows that $\tau([-1, 1]) = 1 - w$. Since $K_m(z) > 1$, normalizing $\nu_m = \tau/(1 - w)$ proves existence and moment equality.

For maximality, suppose a probability measure with the same first $2m$ moments has an atom of mass w' at z and all remaining mass on $[-1, 1]$. Every degree- m polynomial s then satisfies

$$w's(z)^2 \leq \int s^2 d\sigma.$$

Take $s_z(u) = \sum_{j=0}^m \phi_j(z)\phi_j(u)$. Orthonormality gives $s_z(z) = K_m(z)$ and $\int s_z^2 d\sigma = K_m(z)$. Therefore $w' \leq 1/K_m(z)$. $\square$

Lemma 2 (Closed form). *If $A = \operatorname{arcosh}(z)$, then*

$$K_m(z) = m + \frac{1}{2} + \frac{\sinh((2m+1)A)}{2\sinh A}.$$

For every fixed $z > 1$, there are positive constants c_z, C_z such that

$$c_z e^{2mA} \leq K_m(z) \leq C_z e^{2mA}$$

for all $m \geq 1$.

Proof. Since $T_j(z) = \cosh(jA)$,

$$\begin{aligned} K_m(z) &= 1 + 2 \sum_{j=1}^m \cosh^2(jA) \\ &= m + 1 + \sum_{j=1}^m \cosh(2jA) \\ &= m + \frac{1}{2} + \frac{\sinh((2m+1)A)}{2\sinh A}. \end{aligned}$$

The exponential comparison follows because $A > 0$ is fixed. $\square$

For the learning construction, apply the affine map

$$q = \frac{\eta}{2}(u + 1).$$

It sends $[-1, 1]$ to $[0, \eta]$. The point $q = 1 - \eta$ corresponds to

$$z_\eta = \frac{2(1 - \eta)}{\eta} - 1 > 1.$$

Let μ_0 be the image of σ . Lemma 1 gives a probability measure $\tilde{\nu}_m$ supported on $[0, \eta]$ such that

$$\mu_1 = w_m \delta_{1-\eta} + (1 - w_m) \tilde{\nu}_m, \quad w_m = K_m(z_\eta)^{-1}, \quad (2)$$

matches moments through order $2m$ with μ_0 .

B The Single-Source Experiment

Fix $d \geq 1$ and $\alpha \in (0, 1]$. The covariate domain is $\{0, 1, \dots, d\}$ with

$$p_X(0) = 1 - \alpha, \quad p_X(x) = \alpha/d \quad (x \in [d]).$$

Set the bias at coordinate zero to any common value $q_0 \in [0, \eta]$. Under hypothesis $h \in \{0, 1\}$, first draw $q_1, \dots, q_d$ independently from μ_h . Keep this vector fixed, then draw each label from $Y \mid X = x \sim \text{Ber}(q_x)$.

B.1 Poissonization

Let the number of observations be Poisson with mean N . For one light coordinate, write S and T for its zero-label and one-label counts. Given q , Poisson splitting gives

$$S \sim \text{Poi}(\lambda(1 - q)), \quad T \sim \text{Poi}(\lambda q), \quad \lambda = N\alpha/d,$$

independently. The mixed probability is

$$Q_h(s, t) = \frac{e^{-\lambda} \lambda^{s+t}}{s!t!} \mathbb{E}_{q \sim \mu_h} [q^t (1 - q)^s]. \quad (3)$$

Lemma 3 (Poissonized total variation). *For the complete Poissonized observations,*

$$\text{TV}(\mathbf{P}_0^N, \mathbf{P}_1^N) \leq d \mathbb{P}(\text{Poi}(\lambda) \geq 2m + 1).$$

In particular, if

$$N \leq N_0 := \frac{2m + 1}{4e\alpha} d^{\frac{2m}{2m+1}}, \quad (4)$$

then this total variation is at most $1/64$.

Proof. For $s + t \leq 2m$, the expectation in (3) is the expectation of a polynomial of degree at most $2m$. It is equal under the two moment-matching priors. Under either prior, $S + T \sim \text{Poi}(\lambda)$ because its law does not depend on q . Hence the one-coordinate distance is at most $\mathbb{P}(\text{Poi}(\lambda) \geq 2m + 1)$.

After integrating out the independent latent biases, the light-coordinate count pairs are independent. Product subadditivity gives the first claim. If $N \leq N_0$, then

$$\lambda \leq \frac{2m + 1}{4e} d^{-1/(2m+1)}.$$

The Poisson Chernoff bound yields

$$\begin{aligned} d \mathbb{P}(\text{Poi}(\lambda) \geq 2m + 1) &\leq d \left(\frac{e\lambda}{2m + 1} \right)^{2m+1} \\ &\leq 4^{-(2m+1)} \leq \frac{1}{64}. \end{aligned}$$

The heavy-coordinate process is identical under both priors. Conditional on all counts, the ordering of observations is generated by a common Markov kernel. Thus the same bound applies to the full ordered data. $\square$

B.2 A fixed-budget lower bound

Lemma 4 (De-Poissonization). *Assume $N_0 \geq 48$. Every test whose two error probabilities are at most $1/5$ uses more than $N_0/4$ fixed-budget observations.*

Proof. Suppose a test uses at most $n \leq N_0/4$ observations. In a Poisson experiment with mean $2 \max\{n, 12\}$, run the test on the first n observations when they are available. Otherwise output a fair coin. The Poisson mean is at most N_0 . A multiplicative lower-tail bound gives

$$\mathbb{P}(\text{Poi}(2 \max\{n, 12\}) \geq n) \geq 0.9.$$

The Poissonized test has success probability at least

$$\frac{1}{2} + 0.9 \left(\frac{4}{5} - \frac{1}{2} \right) > \frac{3}{4}$$

under both priors. Two distributions that admit such a test have total variation above $1/2$. This contradicts Lemma 3. $\square$

Combining (4) and Lemma 4, every test requires at least

$$L_m := \frac{2m+1}{16e\alpha} d^{\frac{2m}{2m+1}} \quad (5)$$

observations, once this quantity is larger than a universal constant.

C Valid Massart Alternatives

Let $\gamma = 1 - 2\eta$ and set

$$\alpha = \frac{16\epsilon}{\gamma w_m}. \quad (6)$$

The first condition of the main theorem, with a sufficiently small universal constant, ensures $\alpha \leq 1$.

Under the null, every light bias lies in $[0, \eta]$, so the zero function is Bayes optimal. Under the alternative, define

$$H = \sum_{x=1}^d \mathbf{1}\{q_x = 1 - \eta\}.$$

Then $H \sim \text{Bin}(d, w_m)$. Let

$$\mathcal{A} = \{dw_m/2 \leq H \leq 2dw_m\}. \quad (7)$$

Lemma 5 (Gap event). *If $dw_m \geq 64$, then $\mathbb{P}(\mathcal{A}) \geq 0.99$. On $\mathcal{A}$, the Bayes classifier is the indicator of the high coordinates, and the excess error of zero satisfies*

$$8\epsilon \leq \Delta_0 \leq 32\epsilon.$$

Its disagreement probability with Bayes is at most $32\epsilon/\gamma$.

Proof. The two-sided multiplicative Chernoff bound gives

$$\mathbb{P}(\mathcal{A}^c) \leq e^{-dw_m/8} + e^{-dw_m/3} < 0.01.$$

The only points where zero differs from Bayes are the H high coordinates. Every such point has label margin $2(1 - \eta) - 1 = \gamma$. Therefore

$$\Delta_0 = \frac{\alpha \gamma H}{d}.$$

Substituting (6) and the bounds in (7) proves the excess interval. The disagreement mass is $\alpha H/d \leq 2\alpha w_m = 32\epsilon/\gamma$. $\square$

Substitution into (5) gives

$$L_m = \frac{(2m+1)\gamma w_m}{256\epsilon\epsilon} d^{\frac{2m}{2m+1}}. \quad (8)$$

D The Variance-Sensitive Holdout

The next result is stated conditionally on any fixed valid environment and any fixed output predictor. It therefore remains valid after conditioning on the learner’s transcript.

Lemma 6 (Holdout test). *Let f^* be a Bayes classifier satisfying the Massart condition with margin lower bound γ . Let $\hat{f}$ have excess error at most ϵ , and let f_0 be zero. Under a null environment where $f_0 = f^*$, define*

$$D = \text{err}(f_0) - \text{err}(\hat{f}).$$

Then $D \in [-\epsilon, 0]$. Under an alternative in $\mathcal{A}$, $D \geq 7\epsilon$. In both cases, the empirical difference from

$$M = \left\lceil \frac{20 \log(2/\delta_v)}{\gamma \epsilon} \right\rceil \quad (9)$$

fresh samples is within 2ϵ of D with probability at least $1 - \delta_v$. Hence thresholding it at 3ϵ distinguishes the cases.

Proof. For every predictor f , the pointwise Massart identity gives

$$\text{err}(f) - \text{err}(f^*) = \mathbb{E}[(1 - 2\eta^*(X))\mathbf{1}\{f(X) \neq f^*(X)\}],$$

where $\eta^*(X) \leq \eta$. Thus an ϵ -optimal predictor obeys

$$\mathbb{P}(\hat{f}(X) \neq f^*(X)) \leq \epsilon/\gamma. \quad (10)$$

Under the null, $f_0 = f^*$, which proves $D \in [-\epsilon, 0]$. Under an alternative in $\mathcal{A}$,

$$D = \Delta_0 - (\text{err}(\hat{f}) - \text{err}(f^*)) \geq 7\epsilon.$$

For a fresh sample let

$$Z = \mathbf{1}\{f_0(X) \neq Y\} - \mathbf{1}\{\hat{f}(X) \neq Y\}.$$

It vanishes whenever $f_0(X) = \hat{f}(X)$. Under the null, $\text{Var}(Z) \leq \epsilon/\gamma$. Under the alternative, Lemma 5 and (10) give

$$\text{Var}(Z) \leq \mathbb{P}(f_0(X) \neq \hat{f}(X)) \leq 33\epsilon/\gamma.$$

Also $|Z - \mathbb{E}Z| \leq 2$. Bernstein’s inequality yields

$$\mathbb{P}(|\bar{Z} - D| > 2\epsilon) \leq 2 \exp(-M\gamma\epsilon/20) \leq \delta_v.$$

The two claimed intervals are separated by the threshold 3ϵ . $\square$

E Adaptive Multi-Source Lifting

Let the covariate space be the disjoint union of k blocks $\mathcal{X}_1, \dots, \mathcal{X}_k$. Block i has one heavy point $x_{i,0}$ and light points $x_{i,1}, \dots, x_{i,d}$. Define

$$\mathcal{F} = \{f_0\} \cup \bigcup_{i=1}^k \{\mathbf{1}_S : S \subseteq \{x_{i,1}, \dots, x_{i,d}\}\}. \quad (11)$$

Lemma 7 (VC dimension). *The class in (11) has VC dimension exactly d .*

Proof. The d light points in any one block are shattered. A set containing a heavy point cannot be shattered because every function labels that point zero. A set of $d + 1$ light points must span at least two blocks. The labeling that is one on a point from each of two blocks is not realized by $\mathcal{F}$. $\square$

Source i is supported on $\mathcal{X}_i$ with mass $1 - \alpha$ on $x_{i,0}$ and α/d on each light point. Set every heavy bias to a fixed value in $[0, \eta]$.

Under the common null prior Π_0 , independently draw every source’s light bias vector from μ_0^d . Each realization is a valid Massart instance with shared Bayes classifier f_0 . For a fixed i , prior Π_i replaces only source i by μ_1^d . Its high-coordinate indicator is in $\mathcal{F}$ and equals zero on all other supports. It is therefore a shared Bayes classifier for every realization under Π_i .

Theorem 8 (Adaptive lifting). *Suppose a personalized learner succeeds with probability at least 0.9 on every valid instance. If*

$$m\gamma^2 w_m d^{\frac{2m}{2m+1}} \geq C \quad (12)$$

for a sufficiently large universal C , then under the common null

$$\mathbb{E}_{\Pi_0}[T_i] \geq L_m/100$$

for every source i , where T_i is its query count and L_m is given by (8).

Proof. Fix i and suppose instead that $\mathbb{E}_{\Pi_0}[T_i] < L_m/100$. We build a test between the single-source priors from Appendix B. Run the MDL learner. Whenever it queries source i , return a sample from the unknown testing oracle after shifting its covariate into block i . For each other source, independently draw a null bias vector once and simulate samples from that fixed environment.

If the source- i query count reaches $L_m/2$, stop and output alternative. If the learner terminates first, take the source- i output and run the holdout test from Lemma 6 with $\delta_v = 0.02$.

Under the null, Markov's inequality gives

$$\mathbb{P}_{\Pi_0}(T_i \geq L_m/2) < 0.02.$$

The learner succeeds with probability at least 0.9, including after averaging over the valid randomized null. Conditional on its transcript and success, the holdout accepts with probability at least 0.98. A union bound shows that the constructed test accepts the null with probability above 0.8.

Under the alternative, truncation is a correct output. If truncation does not occur, the intersection of $\mathcal{A}$, learner success, and holdout success makes the test output alternative. This intersection has probability at least $0.99 - 0.1 - 0.02 > 0.8$. Pointwise correctness of the learner applies to every alternative realization, not only after conditioning on $\mathcal{A}$.

The unknown-oracle budget is at most $L_m/2 + M$. From (8) and (9), condition (12) with a sufficiently large C ensures $M \leq L_m/4$. It also ensures $N_0 = 4L_m \geq 48$. We have constructed a test using less than L_m observations with both errors below $1/5$, contradicting Lemma 4. Therefore $\mathbb{E}_{\Pi_0}[T_i] \geq L_m/100$. $\square$

Summing Theorem 8 over i under the same null gives

$$\begin{aligned} \mathbb{E}_{\Pi_0} \left[\sum_{i=1}^k T_i \right] &\geq \frac{kL_m}{100} \\ &= \Omega \left(\frac{km\gamma w_m}{\epsilon} d^{\frac{2m}{2m+1}} \right). \end{aligned}$$

Since every realization in the support of Π_0 is valid, some fixed environment has at least this expected cost. This proves the main theorem for expected worst-case sample complexity. A deterministic worst-case sample cap is at least the same quantity.

F Optimization and Comparison

F.1 Fixed exponent loss

Fix $\eta \in (0, 1/2)$ and $\zeta > 0$. Choose a fixed integer m satisfying

$$\frac{1}{2m+1} \leq \zeta.$$

Then $w_m > 0$ is a constant depending only on η and ζ . Choose $\epsilon \leq c\gamma w_m$. For all sufficiently large d , the count and holdout conditions hold. The main theorem gives

$$\Omega_{\eta, \zeta} \left(\frac{k}{\epsilon} d^{2m/(2m+1)} \right) = \Omega_{\eta, \zeta} \left(\frac{k}{\epsilon} d^{1-\zeta} \right).$$

F.2 Polynomial accuracy

Fix $\beta > 0$ and suppose $\epsilon \leq d^{-\beta}$. Write $D = \log d$ and $A = A_\eta > 0$. Choose an integer m such that

$$2m \asymp \sqrt{D/A}. \quad (13)$$

For all sufficiently large d , Lemma 2 gives

$$w_m \geq c_\eta \exp(-2mA).$$

The accuracy condition holds because

$$\log(1/\epsilon) \geq \beta D \gg 2mA + O_\eta(1) = O_\eta(\sqrt{D}).$$

Similarly,

$$dw_m \geq c_\eta \exp(D - O_\eta(\sqrt{D})) \rightarrow \infty.$$

For the holdout condition,

$$\begin{aligned} m\gamma^2 w_m d^{2m/(2m+1)} &\geq c_\eta m \exp\left(D - 2mA - \frac{D}{2m+1}\right) \\ &\geq \exp(D - O_\eta(\sqrt{D})) \rightarrow \infty. \end{aligned}$$

Thus all conditions hold. The main lower bound is at least

$$\frac{c_\eta k d m}{\epsilon} \exp\left(-2mA - \frac{D}{2m+1}\right) \geq \frac{c_{\eta,\beta} k d}{\epsilon} \exp(-C_\eta \sqrt{\log d}).$$

F.3 Separate empirical risk minimization

For completeness, the Massart identity used in (10) also gives the Bernstein condition

$$\text{Var}(\ell_f - \ell_{f^*}) \leq \frac{\text{err}(f) - \text{err}(f^*)}{\gamma}.$$

Standard localized VC empirical risk minimization therefore learns one source to excess ϵ and failure probability δ/k from

$$O\left(\frac{d \log(1/\epsilon) + \log(k/\delta)}{\gamma \epsilon}\right)$$

samples (Boucheron et al., 2005; Massart and Nédélec, 2006). Applying it independently to all sources and taking a union bound gives the upper benchmark stated in the main paper.

References

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